Intro to AI Assignment 2

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Problem Statement:

The objective of this assignment is to optimize the locomotion of a HalfCheetah-v5 robot using an Evolutionary Algorithm. The task involves evolving a sequence of K=5 keyframes, where each keyframe consists of 6 torque values and 1 duration value. The fitness is determined by the cumulative reward provided by the Gymnasium environment, calculated as forward velocity minus a control penalty.

Evolutionary Algorithm Design:

The configuration used for these experiments consists of:

- **Generations:** 100
- **Population Size:** 20
- **Keyframes (K):** 5 (6 torques + 1 duration per frame)
- **Mutation Rate:** 0.1

Experiment 1: Tournament (Parents) + Truncation (Survivors):

Description: This experiment used binary tournament selection for parents and truncation for survivor selection.

Graph Analysis:

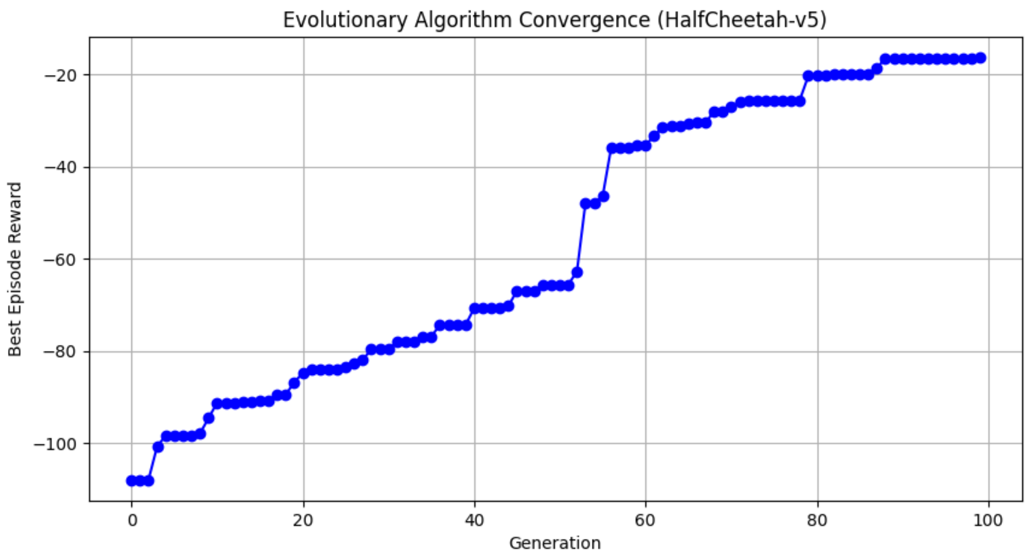

Figure 1: Convergence Plot: Tournament + truncation (100 Generations).

The convergence plot for this experiment shows a monotonic trend. Since truncation strictly preserves the best individuals, the best reward never decreases. The graph displays a staircase pattern with prominent flat areas, indicating that the population frequently settled into local optima before finding a useful mutation. The final reward reached was **-16.42**.

Best Keyframe Data:

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The final evolved gait prioritized stability, as seen by the relatively long durations in the keyframes.

Frame	T1	T2	T3	T4	T5	T6	Duration
1	-0.12	0.07	0.09	0.46	0.49	-0.23	21
2	-0.07	0.14	-0.29	0.20	-0.08	-0.39	27
3	0.45	0.08	-0.08	0.17	0.04	-0.02	27
4	-0.73	-0.01	-0.29	-0.22	-0.31	-0.25	22
5	-0.21	-0.19	0.19	0.05	-0.16	0.37	49

Computational Analysis:

This setup was computationally fast. Survivor selection was performed via truncation. This logic is highly efficient, allowing generations to process nearly instantaneously once evaluations are complete.

Experiment 2: Tournament (Parents) + Tournament (Survivors):

Description: This experiment used binary tournament selection for both parents and survivor selection.

Graph Analysis:

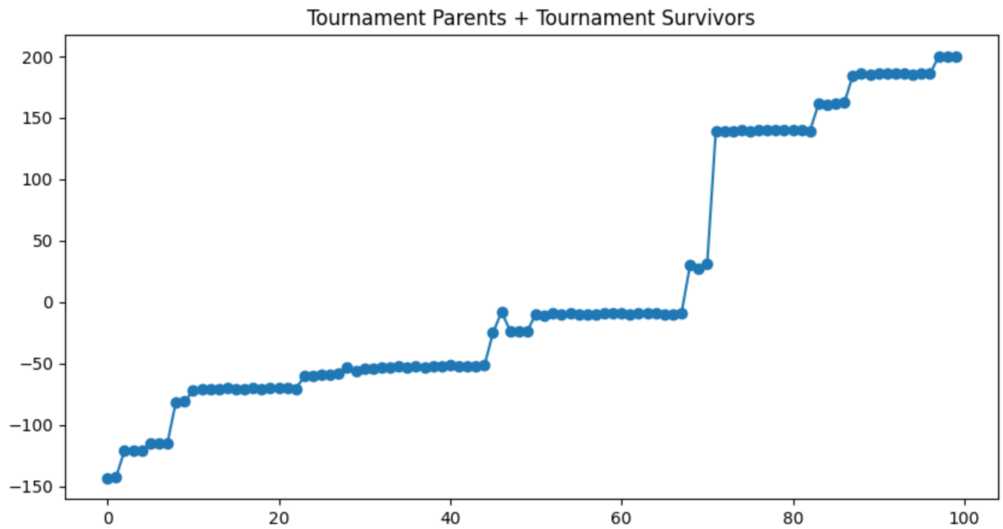

Figure 2: Convergence Plot: Tournament + tournament (100 Generations).

Unlike Experiment 1, this graph is non-monotonic, showing visible dips (e.g., Generations 46–49). This is due to the lack of elitism; as with tournament selection, even the best individual could be deselected if paired against a slightly better one or simply replaced. However, a massive jump occurred at Generation 71, where the reward jumped from -10 to over +139. The final reward achieved was **+200.22**.

Best Keyframe Data:

This gait is high frequency and explosive, characterized by significantly shorter durations compared to the truncation experiment.

Frame	T1	T2	T3	T4	T5	T6	Duration
1	-0.90	-0.89	-0.85	0.49	-0.49	0.31	5
2	-0.17	0.98	-0.03	0.50	0.34	0.02	13
3	-0.01	0.96	0.15	-0.53	-0.85	0.30	33
4	0.34	-0.40	0.33	0.22	0.08	-0.41	7
5	0.21	-0.48	0.04	-0.09	-0.60	-0.55	5

Computational Analysis:

This setup was significantly slower. The survivor selection through Tournament is more complex than Truncation, as it involves a stochastic loop where random pairings must be resolved. Managing these indices and sampling the pool multiple times per generation adds substantial overhead compared to a simple sort command.

Experiment 3: Tournament (Parents) + Fitness Proportional (Survivors):

Description: This experiment used binary tournament selection for parents and Fitness Proportional (Roulette) for survivor selection.

Graph Analysis:

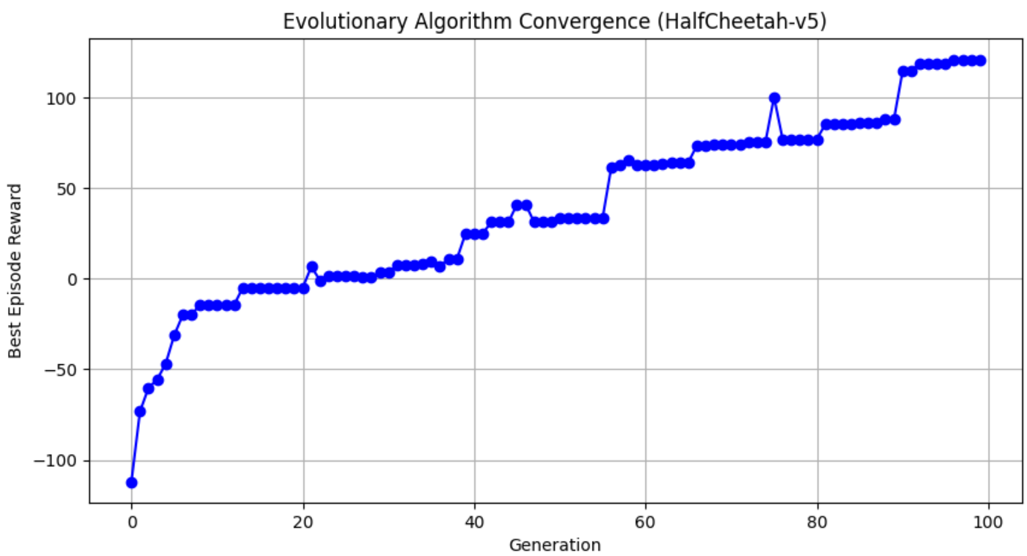

Figure 3: Convergence Plot: Tournament + Roulette (100 Generations).

Similar to Experiment 2, this graph is non-monotonic. Noticeable dips occur throughout the run (e.g., a sharp drop from +100.28 at Generation 75 down to +77.05 at Generation 76). Because this method is strictly probabilistic and lacks elitism, the absolute best individual frequently fails to survive the spin of the wheel. Despite this, the population successfully learned to move forward, crossing into positive rewards permanently around Generation 21 and finishing with a strong final reward of **+120.82**.

Best Keyframe Data:

The evolved gait shows a varied rhythm, utilizing a mix of quick, explosive twitches (durations of 5) followed by longer, stabilizing motions (durations up to 31).

Frame	T1	T2	T3	T4	T5	T6	Duration
1	-0.46	0.97	0.35	0.48	-0.03	0.39	5
2	0.18	0.07	0.15	-0.97	0.09	0.19	24
3	-0.90	-0.77	-0.46	-0.54	0.48	-0.25	5
4	0.15	0.83	0.45	-0.07	0.06	0.23	16
5	0.34	0.35	0.46	0.43	-0.19	0.22	31

Computational Analysis:

This setup was moderately fast, positioning itself computationally between truncation and tournament survivors. The primary overhead stems from the necessary linear transformation (shifting negative rewards) to calculate valid probabilities for the roulette wheel.

Experiment 4: Truncation (Parents) + Tournament (Survivors):

Description: This experiment used truncation selection for parents and binary tournament for survivor selection.

Graph Analysis:

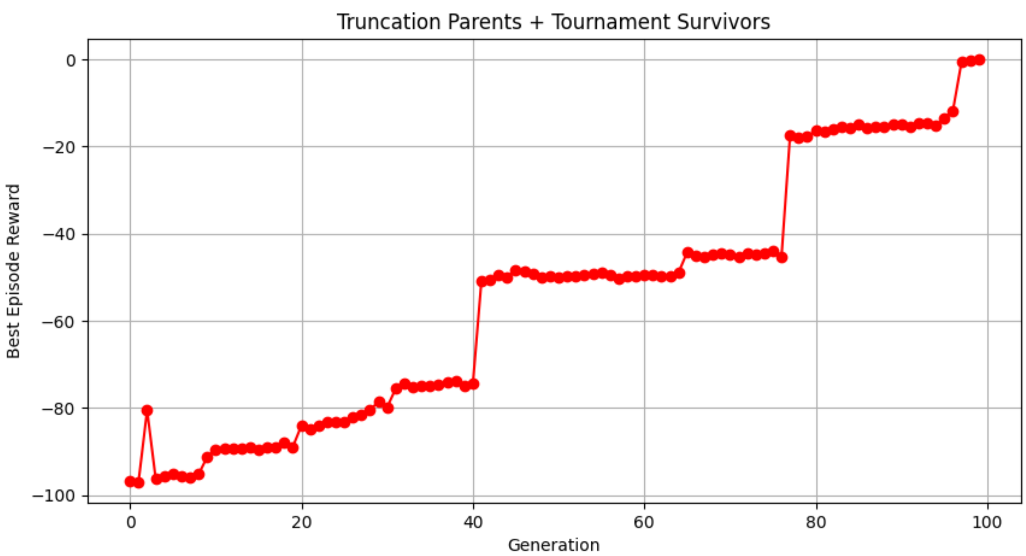

Figure 4: Convergence Plot: Truncation + Tournament (100 Generations).

Unlike the roulette survivor method, this convergence plot displays a staircase pattern and is almost entirely monotonic. Because only the absolute elite are allowed to breed as a result of truncation, the population quickly converges on a specific movement pattern and gets stuck in local optima, resulting in long, flat areas of the graph (for e.g., the population stays on a flat part from Generation 41 to 76). However, because tournament survivor selection preserves a slight amount of genetic diversity, the algorithm eventually gets a good mutation. This results in sudden sharp spikes, most notably the massive jump at Generation 77. The cheetah finished at **-0.09**.

Best Keyframe Data:

The evolved gait relies on longer, sustained joint movements. The hold durations are spread between 14 and 30 frames, indicating a smoothness.

Frame	**T1**	**T2**	**T3**	**T4**	**T5**	**T6**	**Duration**
1	-0.98	-0.47	-0.19	-0.66	0.84	-0.04	30
2	0.38	0.25	-0.25	0.08	0.89	-0.00	18
3	0.81	0.83	-0.05	-0.38	-0.40	-0.47	22
4	0.35	-0.80	0.63	0.33	-0.47	-0.88	14
5	0.30	0.69	0.51	0.77	-0.04	0.72	15

Computational Analysis:

This setup was not is highly efficient. Truncation parent selection only requires a single sorting operation, avoiding the computational overhead of roulette shifting. The Tournament survival phase added some overhead, making this not very fast

Experiment 5: Truncation (Parents) + Truncation (Survivors):

Description: This experiment used truncation selection for both parents and survivors.

Graph Analysis:

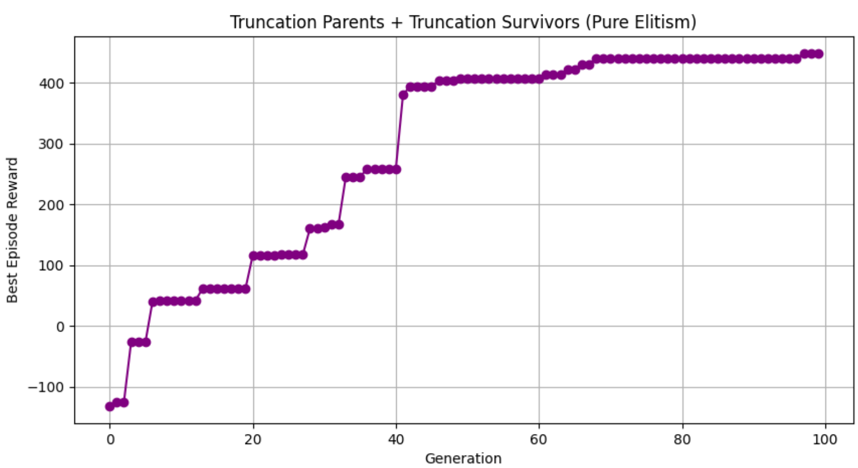

Figure 5: Convergence Plot: Truncation + Truncation (100 Generations).

This convergence plot displays a perfect, rigid staircase pattern and is strictly monotonic. Because only the absolute top performers are allowed to breed and survive, the fitness score never decreases. However, this pure elitism completely destroys genetic diversity, causing the population to get stuck in local optima for long periods (for e.g., there is a massive flat area on the graph from generation 68 all the way to Generation 96 where the reward stays exactly at 440.91). The algorithm only improves when a lucky mutation occurs, leading to aggressive spikes (for e.g., the jump from 258.24 to 380.37 at Generation 41). Despite the flat areas, this aggressive strategy resulted in an exceptionally high final reward of 447.55.

Best Keyframe Data:

Unlike the previous experiment, the evolved gait here relies on short, explosive, and rapid joint movements. The hold durations are very narrow, spreading only between 6 and 11 frames, indicating a fast, twitchy sprinting style rather than a smooth walk.

Frame	**T1**	**T2**	**T3**	**T4**	**T5**	**T6**	**Duration**
1	0.43	-0.41	-0.39	0.29	-0.37	0.07	8
2	0.66	0.03	0.22	-0.10	0.10	-0.75	11
3	0.81	-0.93	-0.79	-0.69	-0.60	0.32	6
4	-0.74	-0.99	-0.05	0.07	0.86	0.78	10
5	0.59	-0.11	0.44	0.28	-0.17	-0.23	11

Computational Analysis:

This setup is highly efficient. truncation for both parents and survivors only requires standard array sorting operations. It completely avoids the computational overhead of roulette shifting or running repetitive while loops for tournament battles. This makes it the fastest configuration to compute per generation.

Experiment 6: Truncation (Parents) + Roulette (Survivors):

Description: This experiment used truncation selection for parents and fitness proportional (roulette) selection for survivors.

Graph Analysis:

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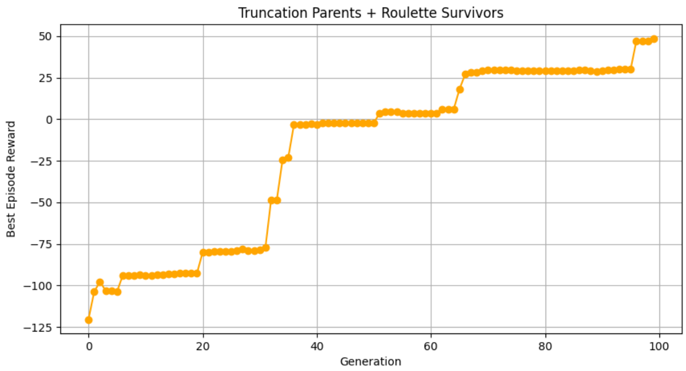

Figure 6: Convergence Plot: Truncation + Roulette (100 Generations).

The convergence plot for this configuration highlights the conflicting nature of the two selection methods. While the overall trend aggressively climbs upwards due to the elite parent selection, the graph is non-monotonic and features many dips. Because the roulette wheel is probabilistic, the absolute best offspring usually fails to survive the spin (for e.g., there is a clear dip from Generation 54, +4.50, down to Generation 55, +3.38, and another drop from Generation 87, +29.76, down to Generation 89 ,+28.86). Despite this, the strict parent breeding ensures the population quickly recovers. The algorithm successfully crossed into positive rewards at Generation 51 and finished with a solid final reward of **+48.76**.

Best Keyframe Data:

The resulting gait for this cheetah is highly asymmetric. It utilizes a combination of very rapid, explosive twitches immediately followed by long, sustained stabilizing poses.

Frame	T1	T2	T3	T4	T5	T6	Duration
1	0.59	-0.02	0.03	-0.25	0.92	0.69	9
2	-0.91	-0.97	-0.21	0.75	-0.71	-0.13	24
3	0.21	0.17	0.33	0.44	0.15	0.49	15
4	-0.89	0.73	0.40	-0.28	-0.84	0.74	8
5	-0.88	-0.59	0.22	-0.28	0.11	-0.42	43

Computational Analysis:

This setup introduces moderate computational overhead. While selecting parents through truncation is highly efficient, the roulette survival phase requires shifting all scores to handle negative values, summing the totals, and calculating probabilities for the entire pool in every generation. This makes it slightly slower than pure truncation or tournament setups, but it successfully preserves genetic diversity.

Experiment 7: Roulette (Parents) + Tournament (Survivors):

Description: This experiment used fitness proportional (roulette) selection for parents and tournament for survivors.

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Graph Analysis:

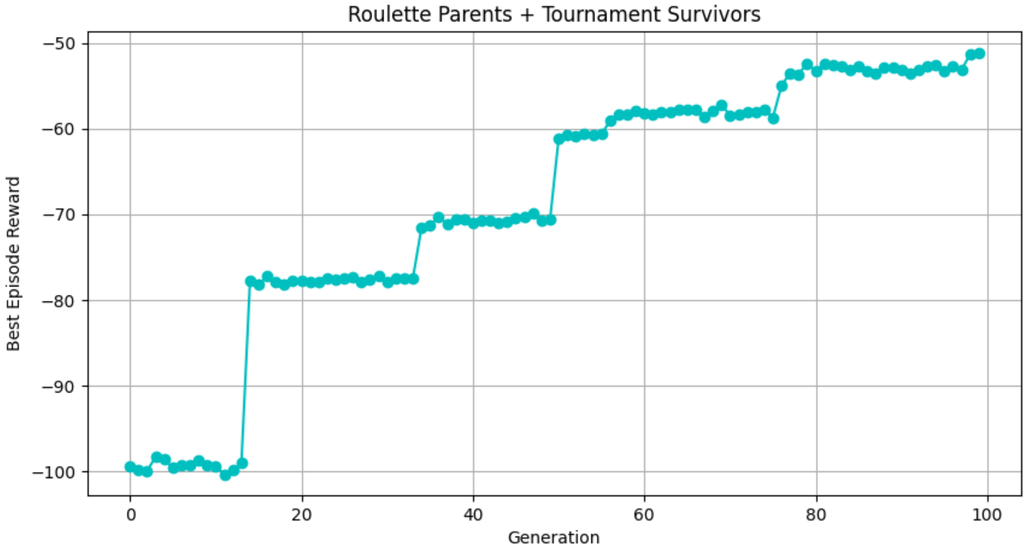

Figure 7: Convergence Plot: Roulette + Tournament (100 Generations).

This convergence plot demonstrates a noisy staircase pattern. Because roulette wheel selection was used for the parents, selection pressure during the breeding phase is very weak. Almost any individual, even those with poor fitness, has a chance to pass on its genes. As a result, the population struggles to use good mutations effectively. The graph shows multiple flat areas (for e.g., from Generation 14 to 33 staying at around -77, and Generation 50 to 75 staying at around -60), but unlike the perfectly flat lines of pure truncation, these flat lines are full of small fluctuations and dips due to the constant introduction of suboptimal genes. While the tournament survivor selection manages to slowly filter out the absolute worst offspring, the overall progress is very slow. The algorithm failed to reach a positive score within 100 generations, finishing with a final reward of **-51.10**.

Best Keyframe Data:

The evolved gait for this configuration heavily favors long, sustained poses with very little rapid movement. With durations of 29, 30, and 34 frames dominating the sequence, the cheetah is making use of a slow, dragging motion rather than an efficient running rhythm.

Frame	**T1**	**T2**	**T3**	**T4**	**T5**	**T6**	**Duration**
1	-0.43	0.82	0.40	0.40	-0.10	-0.14	29
2	0.62	-0.24	-0.32	0.95	-0.35	0.51	6
3	0.23	-0.73	0.01	-0.67	0.55	-0.34	30
4	0.67	-0.65	-0.10	0.00	0.56	0.55	16
5	-0.16	0.04	-0.45	-0.30	-0.35	-0.05	34

Computational Analysis:

This setup is moderately computationally expensive and relatively inefficient. The roulette parent selection requires shifting scores, summing totals, and calculating probabilities for the entire pool to pick parents. Following that, the Tournament survival phase requires executing multiple random comparisons to fill the new population. Given the high overhead and the slow convergence rate this combination proved to be one of the least effective strategies.

Experiment 8: Roulette (Parents) + Truncation (Survivors):

Description: This experiment used fitness proportional (roulette) selection for parents and truncation for survivors.

Graph Analysis:

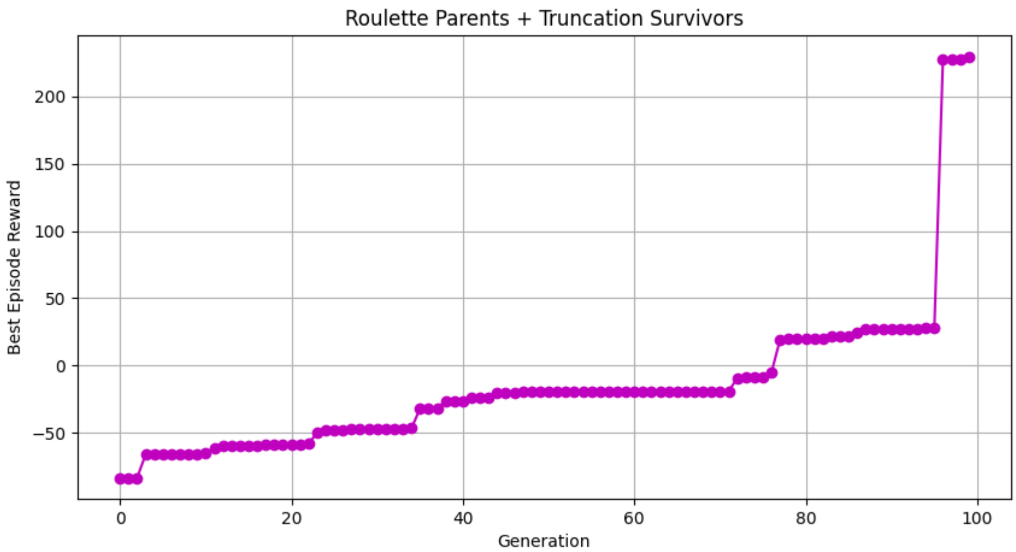

Figure 8: Convergence Plot: Roulette + Truncation (100 Generations).

This convergence plot displays a perfectly monotonic pattern. Because truncation selection is used for the survivors, the absolute best individuals are never discarded, meaning the graph can never go down. However, because the roulette wheel allows suboptimal parents to breed, the algorithm still experiences significant flat areas on the graph where the population struggles to find an immediate improvement (for e.g., the flat period from Generation 54 to Generation 71, staying at around -19.19). The primary advantage of this parent strategy is visible at the very end of the run. The diverse gene pool eventually produced a massive breakthrough, causing the fitness to explode from 28.03 in Generation 95 to 227.30 in Generation 96. The algorithm finished with a very strong final reward of **229.70**.

Best Keyframe Data:

The evolved gait for this cheetah relies heavily on short, rapid joint movements broken up by a single long keyframe. Four of the five keyframes have very narrow hold durations of just 7 or 8 frames, representing explosive twitches, while one keyframe acts as a stabilizer holding for 25 frames.

Frame	**T1**	**T2**	**T3**	**T4**	**T5**	**T6**	**Duration**
1	-0.46	0.76	0.29	-0.69	0.66	-0.26	8
2	0.81	0.20	0.07	-0.49	-0.90	0.45	7

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Frame	T1	T2	T3	T4	T5	T6	Duration
3	-0.79	-0.74	-0.84	0.26	-0.45	0.46	8
4	0.15	0.97	0.20	-0.62	0.27	0.27	25
5	0.40	-0.94	0.42	-0.20	0.33	-0.37	8

Computational Analysis:

This setup balances computational overhead. The roulette parent selection requires the mathematical overhead of shifting scores to handle negative values, summing totals, and calculating probabilities. However, the truncation survival phase is highly efficient, requiring only a single array sort operation to slice the top 20 individuals. It is faster than setups using tournament survival but slightly slower than pure truncation.

Experiment 9: Roulette (Parents) + Roulette (Survivors):

Description: This experiment used fitness proportional (roulette) selection for both parents and survivors.

Graph Analysis:

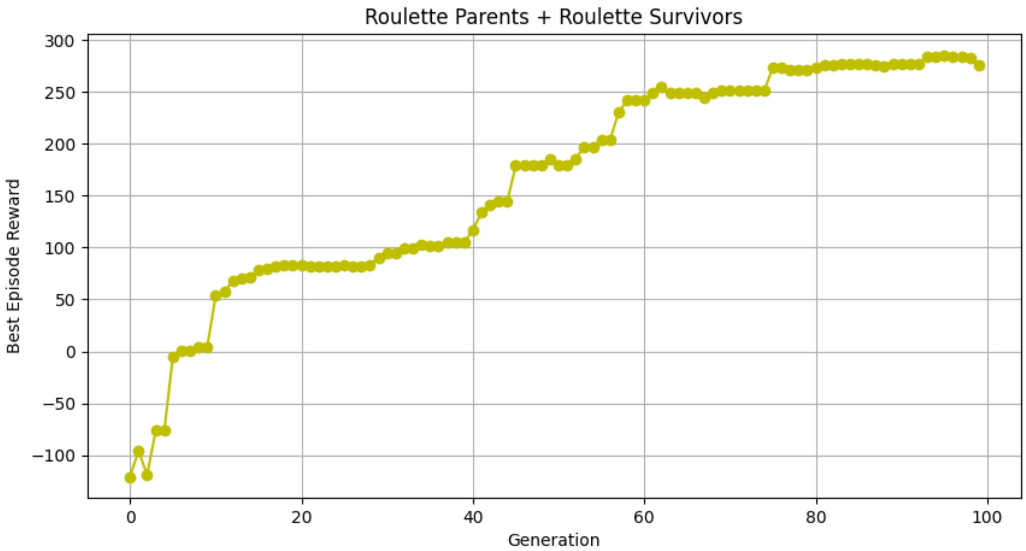

Figure 9: Convergence Plot: Roulette + Roulette (100 Generations).

This convergence plot perfectly illustrates the non-monotonic nature of pure probabilistic selection. Because the roulette wheel is used for both breeding and surviving, there is no guarantee that the best performing cheetah will survive into the next generation. The graph shows a steady overall upward trend but is filled with frequent small dips as good traits are randomly lost and then rediscovered (for e.g., the population reaches a highly impressive peak reward of 285.51 at Generation 95, but because the elite cheetah failed to survive the final spins of the wheel, the best reward actually drops down to 275.78 by Generation 99). Despite this instability, the massive genetic diversity prevented the algorithm from getting permanently stuck in early local optima, allowing it to achieve one of the highest overall scores of the assignment.

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Best Keyframe Data:

The evolved gait for this cheetah is highly rhythmic and dynamic. The hold durations alternate between short, quick snaps and medium length keyframes. This suggests a balanced running pattern rather than the extreme twitching or slow dragging seen in other configurations.

Frame	T1	T2	T3	T4	T5	T6	Duration
1	-0.25	-0.95	0.14	0.26	0.56	0.23	5
2	0.10	-0.27	0.60	0.09	-0.63	-0.04	15
3	0.83	-0.59	-0.38	-0.99	0.42	-0.95	9
4	-0.99	-0.16	-0.33	0.10	0.72	0.74	13
5	-0.78	0.66	-0.35	-1.00	-0.68	-0.89	6

Computational Analysis:

This setup has the highest computational overhead of all the configurations tested. Calculating roulette wheel selections for both the parent phase and the survivor phase requires doing the mathematical heavy lifting twice per generation. For every cycle, the algorithm must find the minimum scores, shift all negative values to positives, sum the total pool, calculate individual probability fractions, and then execute weighted random choices. While it produced a great final runner, it is computationally expensive.

Comparison:

Overall Summary:

This assignment explored the impact of different Evolutionary Algorithm selection strategies on the locomotion of a HalfCheetah-v5 robot. By testing nine different combinations of parent and survivor selection methods (Tournament, Truncation, and Roulette), we observed how the balance between exploration and exploitation directly the final performance of the robot.

Performance Comparison Table

Experiment	Parent Selection	Survivor Selection	Final Reward	Key Characteristic
1	Tournament	Truncation	-16.42	Monotonic staircase, stuck in local optima.
2	Tournament	Tournament	+200.22	Non-monotonic, massive late-stage jump.
3	Tournament	Roulette	+120.82	Strictly probabilistic, frequent dips.
4	Truncation	Tournament	-0.09	Almost entirely monotonic, long flat areas.

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Experiment	Parent Selection	Survivor Selection	Final Reward	Key Characteristic
5	**Truncation**	**Truncation**	**+447.55**	**Highest score, pure elitism, fastest computation**.
6	Truncation	Roulette	+48.76	Elite breeding but probabilistic survival, non-monotonic.
7	Roulette	Tournament	-51.10	Lowest score, weak selection pressure, highly inefficient.
8	Roulette	Truncation	+229.70	Diverse gene pool led to massive late breakthrough.
9	Roulette	Roulette	+275.78	Pure exploration, highly unstable but high performing.

Exploration vs. Exploitation Trade-offs:

- **High Exploitation (The Winner):** Experiment 5 (Truncation + Truncation) achieved the high final reward of +447.55. Because only the absolute top performers were allowed to breed and survive, the fitness score never decreased. However, this pure elitism completely destroyed genetic diversity, causing the algorithm to rely entirely on lucky mutations to escape long local optimas.
- **High Exploration:** On the other end of the spectrum, Experiment 9 (Roulette + Roulette) utilized pure probabilistic selection. While it suffered from instability, where the absolute best individual could fail to survive the spin of the wheel, the massive genetic diversity prevented the algorithm from getting permanently stuck in early local optimas. It reached a peak reward of 285.51and finished at the reward of 275.78.
- **The Worst Performer:** Experiment 7 (Roulette + Tournament) proved to be the least effective strategy, finishing with a final reward of -51.10. The selection pressure during the breeding phase was too weak, meaning almost any individual could pass on its genes, causing the population to struggle to use good mutations effectively.

Computational Efficiency:

The choice of selection method heavily impacted the computational overhead per generation.

- **Fastest:** Truncation proved to be the most highly efficient method. It completely avoids the computational overhead of roulette shifting or running repetitive while loops for tournament battles, relying only on standard array sorting operations.
- **Slowest:** Roulette selection introduced the highest computational overhead of all the configurations tested. It requires doing mathematical heavy lifting: finding minimum scores, shifting negative values to positives, summing the total pool, calculating individual probability fractions, and executing weighted random choices. Experiment 9 applied this twice per generation, making it computationally expensive. Tournament selection fell in the middle, as managing indices and sampling the pool multiple times per generation adds substantial overhead compared to a simple sort command.

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Conclusion:

For optimizing the HalfCheetah-v5 locomotion in exactly 100 generations, pure elitism (truncation + truncation) proved to be the most mathematically efficient and highest scoring approach. However, if the algorithm were to run for thousands of generations, a hybrid approach like roulette + truncation might prove superior, as it preserves a diverse gene pool capable of producing massive performance breakthroughs late in the training cycle.